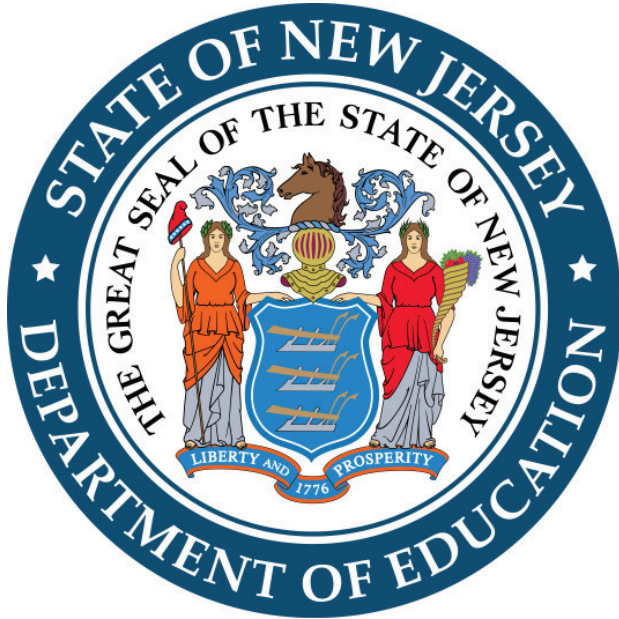




New Jersey
Student Learning
Assessments-Adaptive

Practice Test

Mathematics

Grade 8

Student Name _____



Directions:

Today, you will take the Grade 8 Mathematics Practice Test. You will be able to use a calculator. If you do not know the answer to a question, you may go on to the next question. When you finish, you may review your answers and any questions you did not answer.

1. For questions with bubbled responses, fill in the circle next to your answer choice. If you change your answer, make sure you erase your old answer completely. Do not cross out or make any marks on the other choices.
2. For questions with response boxes, write your answer neatly, clearly, and in the space provided.
3. For questions with response grids, write your answer in the answer boxes at the top of the grid.
 - Write only one digit or symbol in each answer box.
 - Use only the options listed below each box.
 - Be sure to write a negative sign or decimal point in the answer box if it is part of the answer.

4	3							} Answer boxes
-	-	-	-	-	-	-	-	} Negative sign
.	.	.	.	.	.	.	.	} Decimal point
0	0	0	0	0	0	0	0	} Number bubbles
1	1	1	1	1	1	1	1	
2	2	2	2	2	2	2	2	
3	●	3	3	3	3	3	3	
●	4	4	4	4	4	4	4	
5	5	5	5	5	5	5	5	
6	6	6	6	6	6	6	6	
7	7	7	7	7	7	7	7	
8	8	8	8	8	8	8	8	
9	9	9	9	9	9	9	9	



1. An equation is shown.

$$n^2 = 43$$

Select all of the values that represent solutions to this equation.

- (A) -43^2
- (B) $-\sqrt{43}$
- (C) $\sqrt{43}$
- (D) $\left(\frac{1}{2}\right)43$
- (E) 43^2

2. What is $0.8\overline{3}$ written as a fraction?



3. The value of $\sqrt{40}$ is between two consecutive whole numbers.

Complete the inequality to show the two whole numbers.

-	-	-	-	-	-	-
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

 $< \sqrt{40} <$

-	-	-	-	-	-	-
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

4. Select all the tables with data that represent a function.

(A)

x	y
5	5
5	10
10	15
10	20

(C)

x	y
0	0
1	1
2	8
3	27

(E)

x	y
12	6
18	9
24	6
24	4

(B)

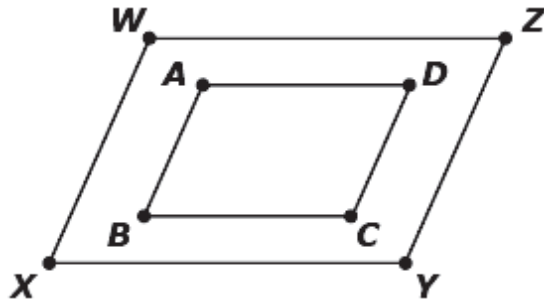
x	y
1	3
2	6
3	9
4	12

(D)

x	y
5	5
15	5
15	10
20	10



5. Parallelograms $ABCD$ and $WXYZ$ are shown.



Karen wants to show that the two parallelograms are similar using transformations.

What transformation must Karen use?

- (A) dilation
- (B) reflection
- (C) rotation
- (D) translation



Mathematics

6. A restaurant manager surveys 80 people in two different age groups on whether they prefer turkey or chicken sandwiches.

Which two-way table shows no association between age and sandwich choice?

(A)

	Turkey	Chicken	Total
People Age 30 and Over	30	10	40
People Under Age 30	10	30	40
Total	40	40	80

(B)

	Turkey	Chicken	Total
People Age 30 and Over	40	0	40
People Under Age 30	0	40	40
Total	40	40	80

(C)

	Turkey	Chicken	Total
People Age 30 and Over	0	40	40
People Under Age 30	40	0	40
Total	40	40	80

(D)

	Turkey	Chicken	Total
People Age 30 and Over	20	20	40
People Under Age 30	20	20	40
Total	40	40	80



7. An inequality is shown, where b is an integer.

$$4.5 < \sqrt{b} < 4.9$$

What is a possible value of b ?

−	−	−	−	−	−	−	−
•	•	•	•	•	•	•	•
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

8. Two cylinders, A and B, are created.

- Cylinder A has volume V .
- Cylinder B has the same height as cylinder A.
- Cylinder B has half the diameter of cylinder A.

Create an expression that represents the volume of cylinder B in terms of V .



9. A system of linear equations has no solutions. One of the equations of the system is shown.

$$y = 2x - 1$$

Which equation could be the other equation of the system?

- (A) $y = -\frac{1}{2}x + 1$
- (B) $y = -2x - 1$
- (C) $y = \frac{1}{2}x + 2$
- (D) $y = 2x + 2$

10. A linear function is modeled in the table as shown.

x	y
4	11
12	13
16	14

A second function has a greater slope and a larger y -intercept than the function modeled in the table.

Which equation could represent the second function?

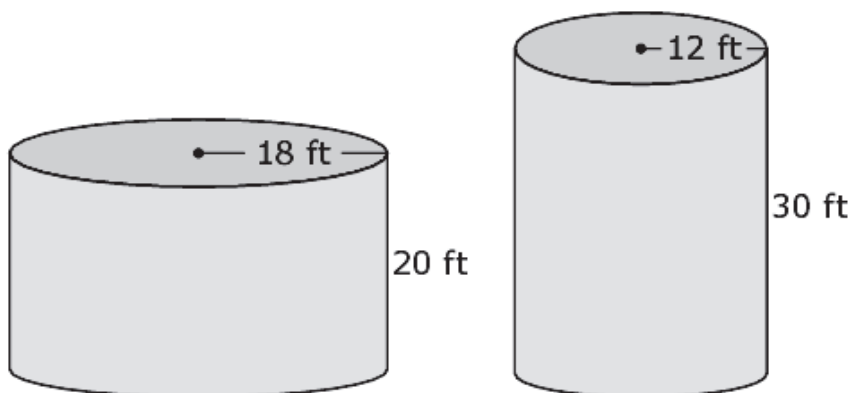
- (A) $y = \frac{1}{4}x + 2$
- (B) $y = \frac{1}{2}x + 6$
- (C) $y = \frac{1}{5}x + 12$
- (D) $y = \frac{1}{2}x + 14$



11. Select whether each equation has no solution, one solution, or infinitely many solutions.

	No solution	One solution	Infinitely many solutions
$7x + 10 = 7x + 10$	(A)	(B)	(C)
$4x = 4x + 3$	(D)	(E)	(F)
$5x + 3 = 2x - 3$	(G)	(H)	(I)
$2x + 8 = 3 + 2x$	(J)	(K)	(L)

12. A farm has two cylindrical silos for storing grain as shown.



How much greater is the volume, in cubic feet, of the larger silo than the smaller silo?



- 13.** Mark recorded the diameter of the same tree trunk every year for 10 years. He modeled the data with the equation shown, where d is the diameter, in inches, of the tree trunk after y years.

$$d = 0.47y + 0.25$$

According to the model, which statement about the tree trunk is true?

- (A) Over the next year, the diameter of the tree trunk will grow exactly 0.25 inch.
- (B) Over the next year, the diameter of the tree trunk will grow exactly 0.47 inch.
- (C) Each year, on average, the diameter of the tree trunk increased by 0.25 inch.
- (D) Each year, on average, the diameter of the tree trunk increased by 0.47 inch.
- 14.** An equation is shown.

$$\left(\frac{1}{6}\right)^p = 36^3 \cdot 6^{18}$$

What is the value of p ?

−	−	−	−	−	−	−
•	•	•	•	•	•	•
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9



- 15.** Kelly is running home from the park. The function shown gives her distance, d , in miles, from home after t minutes.

$$d = -0.15t + 2.3$$

What does -0.15 represent in this function?

- (A) the number of minutes Kelly has been running
 - (B) the total distance from home after each minute
 - (C) the total distance from home when Kelly began running
 - (D) the rate at which the distance from home changes each minute
- 16.** Point R has coordinates (a, b) . The point is reflected across the x -axis and then translated 5 points to the right to create point S .

Create an expression that represents the y -coordinate of S .



- 17.** A system of equations is shown.

$$2x - y = 15$$

$$y = 9$$

What is the value of x in the solution to this system?

-	-	-	-	-	-	-
.	.	.	.	.	.	.
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

- 18.** Charlie and Susan are planning a party for 10 people. Charlie finds a location that charges an initial fee of \$20 plus \$25 per person.

Susan finds a location whose rental fee is represented by the equation $y = 15x + 100$, where x is the number of people in attendance and y is the total cost.

Select all the statements that are true.

- (A) Charlie's location is a cheaper location.
- (B) Susan's location is cheaper for 10 people.
- (C) The charge for each additional person is greater for Susan's location.
- (D) The charge for each additional person is greater for Charlie's location.
- (E) If the number of people at the party changes to 12, the total cost at each location is the same.



Paper Practice Test Answer and Alignment Document: Math

Grade 8 Mathematics

Item Sequence	Answer Key	Standards Alignment
1	B, C	8.EE.A.2
2	$\frac{5}{6}$, or any equivalent fraction	8.NS.A.1
3	6 and 7, or any equivalent values	8.NS.A.2
4	B, C	8.F.A.1
5	A	8.G.A.4
6	D	8.SP.A.4
7	21, 22, 23, or 24, or any equivalent values	8.NS.A.2
8	$\frac{1}{4}V$, or any equivalent expression	8.G.C.9
9	D	8.EE.C.8b
10	D	8.F.A.2
11	Row 1: Infinitely many solutions Row 2: No solution Row 3: One solution Row 4: No solution	8.EE.C.7a
12	2160π , or any value from 6782 to 6789, inclusive	8.G.C.9
13	D	8.SP.A.3
14	-24, or any equivalent value	8.EE.A.1
15	D	8.F.B.4
16	$-b$, or any equivalent expression	8.G.A.3
17	12, or any equivalent value	8.EE.C.8b
18	B, D	8.F.A.2